

Paper 1 MCQ: Answers

1.	A	2.	D	3.	A	4.	C	5.	B
6.	B	7.	A	8.	C	9.	C	10.	A
11.	D	12.	D	13.	A	14.	D	15.	C
16.	A	17.	D	18.	A	19.	B	20.	C
21.	A	22.	A	23.	B	24.	C	25.	D
26.	C	27.	D	28.	B	29.	B	30.	A

Section A: Structured Questions

Question 1 [10 marks]

- (a) Fig 1.1 below shows the results of an investigation into the effect of a competitive inhibitor on an enzyme-controlled reaction over a large range of substrate concentrations.

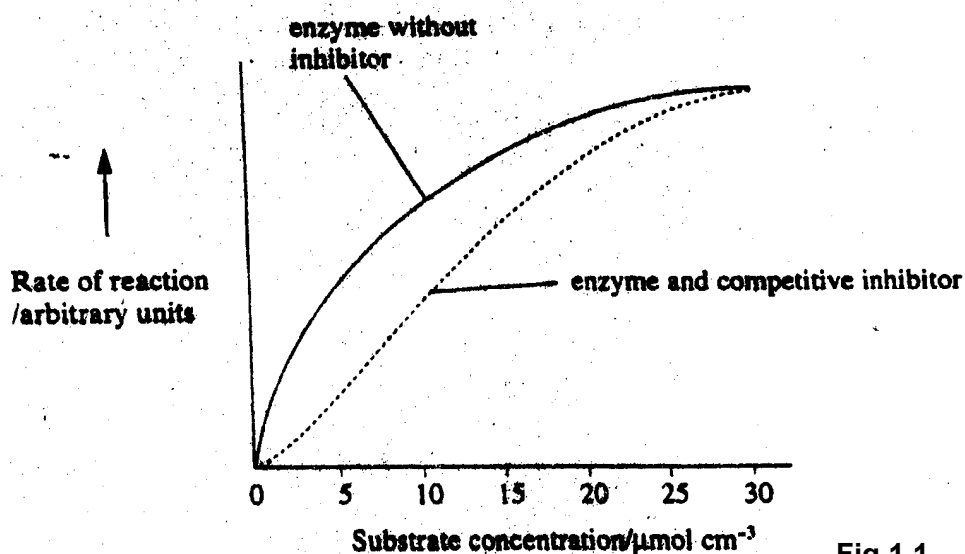


Fig 1.1

- (i) Explain the difference in the rates of reaction between the two experimental conditions at the substrate concentration of $10 \mu\text{mol cm}^{-3}$. [3]
- Rate of reaction without inhibitor is higher than rate of reaction with competitive inhibitor;
 - Inhibitor and substrate have similar shape;
 - Compete for the same active site of enzyme;
 - Low/ $10 \mu\text{mol cm}^{-3}$ substrate concentration decreases probability of effective collision between enzyme and substrate;
 - Less enzyme-substrate complex formed than enzyme-inhibitor complex;
 - Less products formed per unit time thus lower rate of reaction
- (ii) Explain why the rates of reaction at the two experimental conditions are similar at the substrate concentration of $30 \mu\text{mol cm}^{-3}$. [2]

- **High/30 umol cm^{-3} substrate concentration increases probability of effective collision between enzyme and substrate;**
- **More enzyme-substrate complex formed;**
- **More products formed per unit time**
- **Increased rate of reaction**

Alcohol dehydrogenase is an enzyme found in the liver. It normally breaks down ethanol ($\text{C}_2\text{H}_5\text{OH}$) into less harmful products. About 50 deaths each year occur following ingestion of a compound called ethylene glycol ($\text{C}_2\text{H}_4(\text{OH})_2$) which is found in antifreeze. This compound is not lethal, but it is broken down by alcohol dehydrogenase into highly toxic oxalic acid. Administering a large dose of ethanol as quickly as possible can treat ethylene glycol poisoning.

(b) Use your knowledge of enzyme activity to explain how this treatment may counteract the effects of ethylene glycol poisoning. [2]

- **Competitive inhibition;**
- **Ethanol and ethylene glycol have similar shape and compete for the same active site of alcohol dehydrogenase;**
- **Large dose of ethanol increases chance of effective collision between ethanol and enzyme;**
- **More ethanol-enzyme complex formed instead of ethylene glycol-enzyme complex;**
- **Prevents breakdown of ethylene glycol and build-up of oxalic acid;**

(c) Explain what causes a particular enzyme molecule to always fold into the same shape. [3]

- **an enzyme is a globular protein molecule;**
- **with a specific primary structure that folds to form a secondary, then tertiary structure;**
- **globular structure is maintained by bonds between R-groups of amino acids;**
- **ionic, hydrogen, disulphide bonds and hydrophobic interactions;**
- **a particular enzyme molecule will always have the same type of amino acids;**
- **so bonds between R-groups of these amino acids will always be in the same position as determined by the type and position of the R-groups ;**

Question 2 [12 marks]

(a) During embryo development, the cells start to differentiate after the blastula stage.

Describe a DNA modification which enables the cells to differentiate. [2]

DNA methylation;

Addition of methyl group to cytosine;

DNA methyltransferase;

Transcription factors and hence RNA polymerase unable to bind to DNA;

Gene is silenced/not expressed;

(b) Protein synthesis is necessary for the survival of cells where important proteins are synthesised. Protein synthesis involves both the process of transcription and translation. Any disruption to these processes may cause the cell to be dysfunctional.

(i) Describe the importance of the DNA-binding domain and protein-binding domain of transcription factors. [2]

**DNA-binding domain recognizes and;
binds to specific DNA sequences;
Protein-binding domain contains binding sites;
For other transcription factors or proteins (eg RNA polymerase) to bind to;
To form transcription initiation complexes;
importance – initiation / regulation of transcription;**

(ii) A retrotransposon (a 'jumping gene' containing non-coding DNA) was accidentally inserted at the TATA region in the promoter of a gene. Explain how this would affect transcription of that gene. [2]

**change in sequence of bases at TATA region;
DNA binding domain of transcription factor cannot recognize the TATA region;
unable to form hydrogen bonds;
other transcription factors and RNA polymerase cannot bind to promoter;
transcription is unable to occur;**

(iii) State precisely where in the cell translation occurs. [1]

**Ribosomes;
in cytoplasm/ RER;
R: cytoplasm only**

(iv) Describe how the structure of the ribosome is adapted for translation. [3]

**Large subunit contains peptidyl transferase;
for formation of peptide bonds in between amino acids;
Holds the translation initiation complex in place;
contains Peptidyl, aminoacyl and exit sites;
Peptidyl site contains the tRNA with the polypeptide chain attached;
Aminoacyl site is for binding of the new aminoacyl tRNA complex;
Exit site for tRNA to be released from ribosome;
Small subunit for attachment of mRNA;**

(v) A mutation was found in a gene that codes for the capping enzyme complex. Explain how this mutation would affect translation. [2]

capping enzyme complex is non-functional;

**EITHER
capping enzyme complex cannot synthesise 5' cap for pre-mRNA;
mRNA cannot exit the nucleus to ribosomes;
for translation;**

OR

incorrect 5' cap formed;
mRNA cannot bind to small ribosomal subunit;
translation is unable to occur;

Question 3 [10 marks]

- (a) Coat colour in cats is determined by a sex-linked gene with two alleles, black and orange. When black cats are mated with orange cats, the female offspring are always tortoiseshell, their coats show black and orange patches of various sizes, while the male offspring have the same coat colour as their mothers.

Using the appropriate symbols, draw genetic diagrams in the space below to account for this cross. [4]

Let X^B be the allele for black

Let X^G be the allele for orange [1/2 m]

Parental phenotype	Black female cat	X	Orange male cat	[1/2 m]
Parental genotype	$X^B X^B$	X	$X^G Y$	[1/2 m]
Gametes	X^B	X^G	Y	[1/2 m]
F1 genotype	$X^B X^G$		$X^B Y$	[1/2 m]
F1 phenotype	Tortoiseshell female		Black male	[1/2 m]
F1 phenotypic ratio	1	:	1	[1 m]

- (b) A cross was made between two fruit flies. One parent had an unknown genotype since it had the wild phenotype (normal [long] wings and normal body colour). The other parent had vestigial [underdeveloped] wings and ebony body colour, so it was known to be homozygous recessive for both characteristics.

The resulting offspring were as follows:

Normal wings and normal body colour	45
Normal wings and ebony body colour	48
Vestigial wings and normal body colour	44
Vestigial wings and ebony body colour	51

Using the following symbols:

N normal wings	n vestigial wings
E normal body	e ebony body

(i) Draw a genetic diagram in the space below to show the cross described. [4]

Parental phenotype	Wild phenotype	X	Vestigial wings and ebony body		
Parental genotype	NnEe	X	nnee [1 m]		
Gametes	NE nE Ne ne		ne [1/2 m]		
F1 genotype	NnEe	nnEe	Nnee	nnee	[1 m]
F1 phenotype	Normal wings Normal body	Vestigial wings Normal body	Normal wings Ebony body	Vestigial wings Ebony body	[1/2 m]
F1 phenotypic ratio	1	:	1	:	1 : 1 [1 m]

(ii) Vestigial wing and ebony body are caused by artificially induced mutations in the laboratory bred fruit flies. Suggest why such mutations are unlikely to be found in natural populations. [2]

- o **Vestigial wings and ebony body are selected against in the wild/natural populations;**
- o **as vestigial wings are too underdeveloped for flight to escape predators or look for food;**
- o **ebony body is too obvious for predators to detect;**
- o **hence unable to survive and pass on these alleles to the next generation;**
- o **these traits are recessive – requires 2 alleles to express;**

2 max

Question 4 [8 marks]

Fig. 4.1 below shows the main stages in the polymerase chain reaction (PCR).

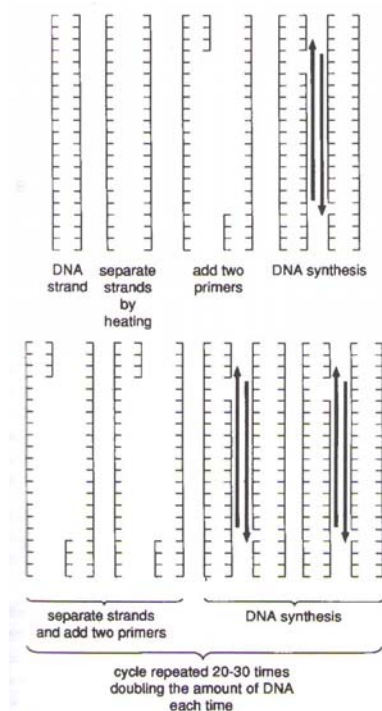


Fig 4.1

(a) Suggest why the brief heat treatment separates two strands of DNA double helix. [2]

High heat – increase kinetic energy;
Increases molecular vibrations;
Breaks hydrogen bonds;
In between bases of the DNA double helix;

(b) Describe the structure and function of a primer in PCR. [2]

Primer – short sequence of DNA;
Single stranded;
Complementary to flanking sequence of target DNA;
For the addition of deoxyribonucleotides to the 3' end;
Taq polymerase cannot add nucleotides from scratch;

(c) Explain the advantages of using Taq DNA polymerase instead of DNA polymerase. [2]

Taq DNA polymerase is able to withstand high temperatures;
Will not denature;
Suitable as heating is required to 95 °C in each cycle for the DNA strands to separate;
Under which DNA polymerase will denature;

Eliminate the need for adding new enzymes into the PCR reaction during the thermocycling process;
Allow automation of process;
2 max

(d) The copying error rate for PCR is much higher as compared to DNA replication *in vivo*. Suggest why this is so. [2]

No exonuclease found in *Taq* polymerase;
Hence unable to proof-read and replace the mismatched deoxyribonucleotides during elongation;

OR

DNA polymerase has 3' to 5' exonuclease;
Able to remove and replace mismatched deoxyribonucleotides;

1 mark for each point

Question 5 [10 marks]

(a) Fig 5.1 below is an electron micrograph of a mesophyll cell from a leaf.

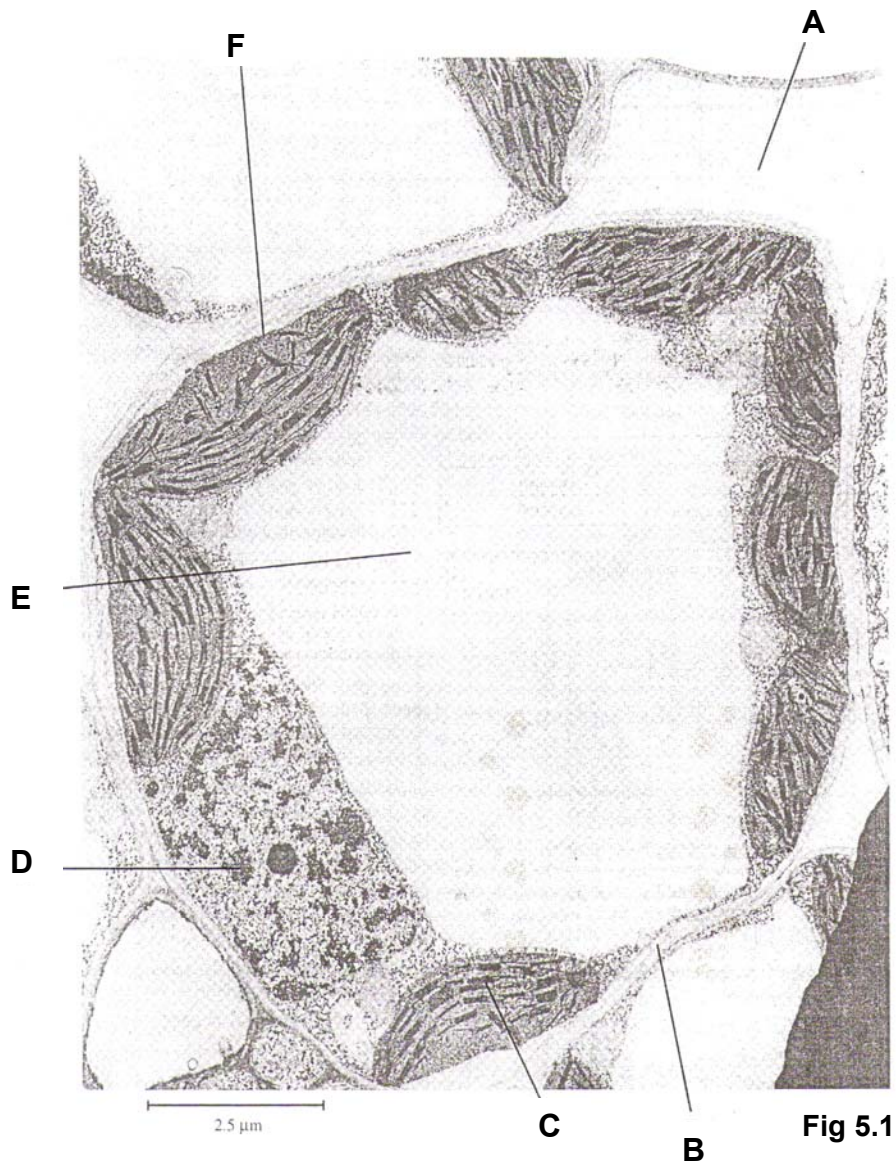


Fig 5.1

(i) Identify, by using the letters A to F, a part of the cell shown in Fig 5.1 where the following substances are located. [2]

chlorophyll **C**

cellulose **B**

DNA **D**

phospholipids **F**

- (ii) State precisely where the light independent reactions (Calvin cycle) occur in plant cells. [1]

Stroma in chloroplast

The pathway of carbon during the reactions of photosynthesis was discovered using the unicellular green algae *Chlorella*. The *Chlorella* culture was supplied with $^{14}\text{CO}_2$ and illuminated with varying length of time. Some algae were removed at intervals and immediately killed by boiling in alcohol. This provided an extract which was examined using chromatographic techniques and any radioactive compounds present showed up on X-ray film as blackened areas.

After 15 seconds of illumination, the extract produced three prominent blackened areas which were due to the presence of three compounds:

- i. glycerate phosphate (GP/ PGA)
- ii. ribulose biphosphate (RuBP)
- iii. triose phosphate (TP/ PGAL)

- (b) In a second series of experiments the algae were illuminated in the presence of $^{14}\text{CO}_2$ for 15 seconds. Illumination of the algae continues for a further 1 minute after all $^{14}\text{CO}_2$ present had been removed. Some algae were then removed, killed and the extract examined. A distinct blackened area was identified on the X-ray film.

Name the compound that was found in this blackened area. Explain your answer. [2]

- **Ribulose biphosphate (RuBP):**
- **In the absence of CO_2 , RuBP is not utilised in carboxylation to form glycerate phosphate (GP/ PGA);**
- **However, in the presence of light, light reaction allows the formation of ATP and NADPH;**
- **Labelled PGA/GP can be reduced to triose phosphate (TP)/G3P;**
- **which in turn is used in the regeneration of RuBP, thus, accumulation of RuBP;**

@ $\frac{1}{2}$, max 2

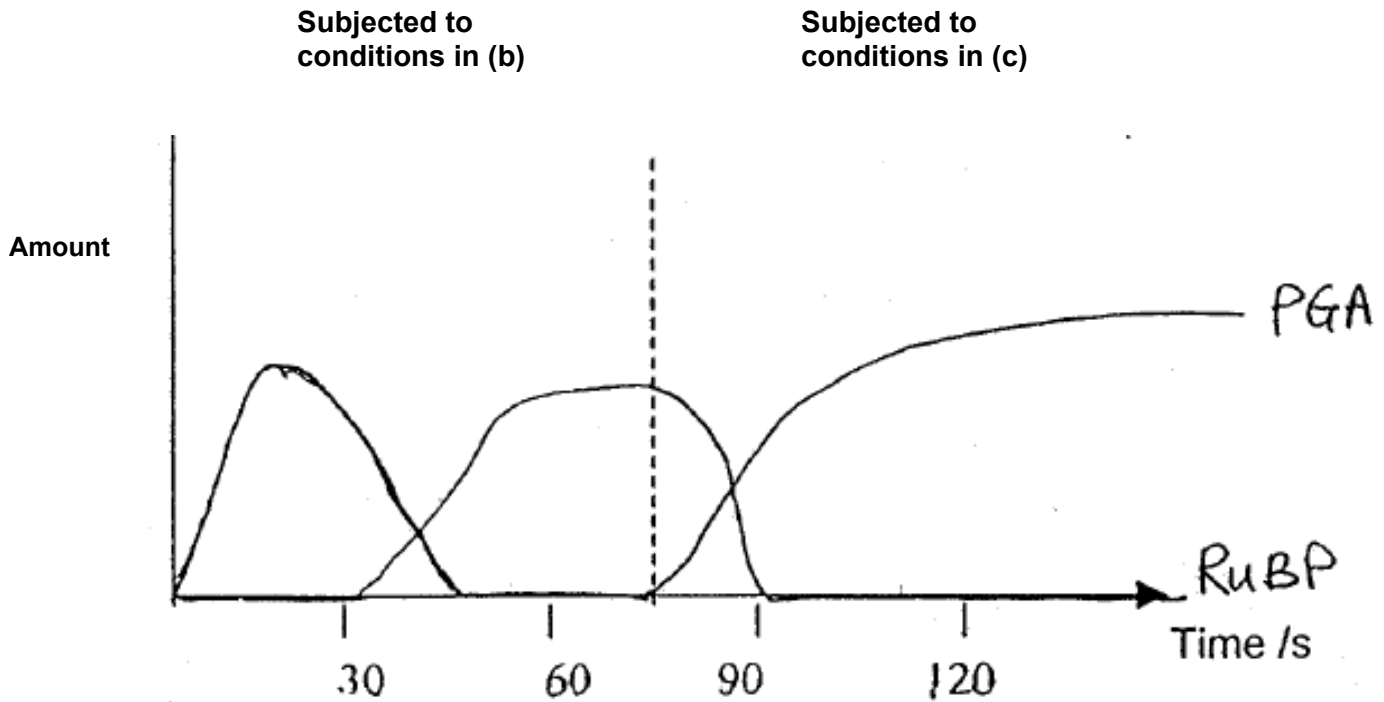
- (c) In another series of experiments, the algae were illuminated in the presence of $^{14}\text{CO}_2$ for 15 seconds and then placed in the dark for 1 min in the presence of $^{14}\text{CO}_2$. Some algae were removed, killed and the extract examined. A distinct blackened area was identified on the X-ray film.

Name the compound that was found in this blackened area. Explain your answer. [2]

- **Glycerate phosphate (GP/ PGA);**
- **CO_2 combines with RuBP to form GP/PGA in the absence of light;**
- **However, in the absence of light, light reaction cannot proceed, thus, ATP and NADPH cannot be made;**
- **GP/PGA cannot be reduced to TP/G3P**
- **and RuBP cannot be regenerated;**

@ $\frac{1}{2}$, max 2

- (d) On the graph below, show how the amounts of the labelled compounds named in (b) and (c) will change if the algae is *first* subjected to the conditions in (b) *then* to conditions in (c). [3]



Overall shape – 1mark
Correct timing -1 mark
Amount of product – 1 mark

Section B
Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate. Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

Question 6

(a) Describe, using examples, the roles of lipids in organisms. [10]

Triglyceride is an important lipid in the biological system of animals:

1. **Supply essential fatty acids needed for proper growth and development of animals;**
2. **Acts as a natural buffer that absorbs shock and cushion fragile internal organs, eg the heart and kidneys from impact;**
3. **Reduce loss of body heat by acting as a thermal insulator below the skin of arctic animals such as whales in the form of blubber;**
4. **less dense than water- provides buoyancy which is important for keeping aquatic animals afloat.**
5. **important source of energy - oxidized to give twice the amount of energy as the same mass of carbohydrate;**
6. **Oxidation of triglycerides during respiration produces metabolic water which is important for desert animals where water is scarce.**
7. **Solvent for fat soluble vitamins eg vit A**

Phospholipids are lipids that contain a phosphate group:

8. **Phospholipid bilayer forms the basic structure of membranes;**
9. **Amphiphatic nature confers selectively permeability to cell, regulate movement of substances across membranes**
10. **unsaturated hydrocarbon tails of phospholipids increases fluidity of plasma membrane**
11. **Fluidity of membrane also allows exocytosis and endocytosis to take place;**
12. **Modified phospholipids such as glycolipids help in cell-cell recognition and adhesion;**
13. **Lipids of myelin sheath aids in the rapid transmission of impulses along myelinated neurons.**

Lipid derivatives:

14. **Lipid derivatives like cholesterol increases fluidity of cell membranes by disrupting the close packing of phospholipids molecules;**
15. **Important for organisms living at cold temperatures to prevent membranes from solidifying;**

10 max

(b) Explain why lipids are better storage molecules than carbohydrates. [7]

1. **Contains a greater proportion of hydrogen to carbon atoms per gram compared to carbohydrates;**
2. **Yields twice the amount of energy per gram when oxidized compared to carbohydrates;**
3. **Has about half the mass, compared to carbohydrates with the same amount of energy stored;**
4. **Useful for animals in which locomotion requires body mass to be kept to a minimum**
5. **And in plants whose seeds are dispersed by wind**
6. **Releases twice the amount of water when oxidized during respiration, compared to same mass of carbohydrates**
7. **Important source of metabolic water for desert animals like camels, kangaroo rats, which store fats for this purpose;**
8. **Less dense than water whereas carbohydrates are not, provides aquatic animals with buoyancy**
9. **Long-term energy store, oxidized after carbohydrates are oxidized, important to hibernating animals**

7 max

(c) How do lipids differ structurally from carbohydrates? [3]

Lipids	Carbohydrates
Contains ester linkages	Contains glycosidic bonds for di- and poly-saccharides
Contains higher proportion of hydrogen to oxygen ratio than carbohydrates	Contains less hydrogen to oxygen ratio than lipids
Esters of fatty acids and an alcohol	Aldehydes or ketones, contain many hydroxyl groups
Generally much smaller molecules	May be large macromolecules eg polysaccharides

3 max

Question 7

Describe the process of obtaining a named protein product using genetic engineering.
[10]

Stage 1:

1. Isolate mRNA for proinsulin/HGH/anti-thrombin from pancreas / β cells of Islets of Langerhans/cells
2. forms cDNA using reverse transcriptase
3. linker DNA added to cDNA

Stage 2:

4. The vector is a plasmid DNA
5. Use of same restriction enzyme (e.g. EcoR1) to cut at restriction site of both vector and DNA containing gene of interest
6. To produce vector with sticky ends that are complementary to the base sequence of the gene of interest
7. Mix to allow annealing of DNA and plasmid vector
8. By complementary base pairing/hydrogen bonding between base pairs
9. Use of DNA ligase to form phosphodiester bonds between the sugar-phosphate backbone
10. The resulting molecule formed is known as recombinant DNA

Stage 3:

11. Recombinant DNA is mixed with bacteria /host cells
12. Host cells take up plasmids by heat shock/electroporation,
13. and become transformed

Stage 4:

14. Bacterial cells grown on nutrient agar plate
15. cells transferred to antibiotic selection medium
16. Cells with plasmid survive, cells without plasmids will die
17. Bacteria cells are then grown in medium containing another antibiotic – cells with reannealed plasmids will survive while those with recombinant plasmids will die

OR

medium containing a colourless compound called X- gal – cell with reannealed plasmids will break down colour X-gal to a blue compound while those with recombinant plasmids will be white.

18. Due to insertional inactivation – gene of interest incorporated in a marker gene causing inactivation of the gene

Stage 5:

19. Colony containing transformed cells are grown in fermenter.

Protein is extracted in the following ways for various proteins:

Insulin

- 20. bacteria cells are burst and proinsulin is extracted**
- 21. Insulin consists of two polypeptide chains (A and B), linked by disulphide bridges, whereas proinsulin consists of three polypeptide chains (A, B and C).**
- 22. Enzymes are used to cut out the C region of this molecule and to combine A and B chains in the correct configuration**

HGH

- 20. Before adding it to the vector, the cDNA has another piece of DNA from the E. coli is added to it.**
- 21. This is the code for a “signal sequence”.**
- 22. When translated, this becomes a sequence of amino acids which when added to the HGH acts like a key to allow the HGH to be secreted across the cell surface membrane and out of the cell.**

Anti-thrombin

- 20. Before adding it to the vector, the cDNA has another piece of DNA added to it.**
- 21. This DNA codes for a peptide that is capable of directing secretion of the polypeptide from the host cells**

Describe the goals, benefits and ethical concerns of the Human Genome Project. [10]

Goals and benefits (each point with example, max 5)

1. Map and sequence the human genome

- **Build genetic and physical maps spanning the human genome.**
- **Determine the sequence of the estimated 3 billion letters of human DNA, to greater than 99.99 percent accuracy.**
- **Chart variations in DNA spelling among human beings.**
- **Map all the human genes.**
- **Begin to label the functions of genes and other parts of the genome.**

2. Map and sequence the genomes of important model organisms (giving examples).

- **Comparison of the complete genome sequences of different organisms enables the tracing of evolutionary relationships among genes and organisms**
- **Discover genes and predict their functions**
- **Assigning functions to genes often involves comparing their sequences with related sequences from model organisms**

3. Collect and distribute data

- **Distribute genomic information and the tools for using it to the research community.**

- Release within 24 hours all sequence data that spans more than 2000 base pairs.
- Create and run databases.
- Develop software for large-scale DNA analysis.
- Develop tools for comparing and interpreting genomic information.
- Share information with the wider public.

4. Study the ethical, legal, and social implications of genetic research

5. Train researchers

6. Develop technologies to make large-scale sequencing faster and cheaper.

- Develop technologies for finding sequence variations.
- Develop ways to study functions of genes on a genomic scale.

7. Transfer technologies to the private sector

Ethical issues (each point with example, max 5)

1. Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others.

- insurers and employers have used the information to deny workers access to health care and jobs

2. Privacy and confidentiality of genetic information.

- usage of genetic information can be used indiscriminately. Slippery slope in the extent genetic information is made available.

3. Psychological impact and stigmatization due to an individual's genetic differences.

- bigoted views about different racial and ethnic groups
- discrimination against individuals carrying certain undesirable genes

4. Reproductive issues including adequate informed consent for complex and potentially controversial procedures, use of genetic information in reproductive decision making, and reproductive rights.

- For example, pre-natal genetic testing can be carried out to screen for the presence of genetic disorders. Parents might then decide to keep or abort the foetus. This leads to a question if it is considered “murder”, since a foetus is a living organism.

5. Clinical issues including the education of doctors and other health service providers, patients, and the general public in genetic capabilities, scientific

limitations, and social risks; and implementation of standards and quality-control measures in testing procedures.

6. Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions.

7. Conceptual and philosophical implications regarding human responsibility, free will vs genetic determinism, and concepts of health and disease.

8. Health and environmental issues concerning genetically modified foods (GM) and microbes.

9. Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials.

10. Eugenics – the use of knowledge of genes to create a superior race